Trends in
Cognitive Sciences

CellPress

Opinion

Can only meat machines be conscious?

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Computational functionalism claims that executing certain computations is sufficient for consciousness, regardless of the physical mechanisms implementing those computations. This view neglects a compelling alternative: that subcomputational biological mechanisms, which realize computational processes, are necessary for consciousness. By contrasting computational roles with their subcomputational biological realizers, I show that there is a systematic tension in our criteria for consciousness: prioritizing computational roles favors consciousness in AI, while prioritizing subcomputational biological realizers favors consciousness in simpler animals. Current theories of consciousness are 'meat-neutral', but if specific physical substrates are necessary, AI may never achieve consciousness. Understanding whether consciousness depends on computational roles, biological realizers, or both, is crucial for assessing the prospects of consciousness in AI and less complex animals.

Highlights

We must recognize that AI and simple animals are competitors when it comes to attribution of consciousness.

If we extrapolate from humans to non-humans to decide whether non-humans are conscious, we must decide whether to extrapolate on the basis of our computational properties or our subcomputational realizers of those properties (or both). The first favors conscious AI over conscious insects, the second the reverse.

In the evolution of animals, purely electrical nervous systems did not lead to animals that are candidates for consciousness, but electrochemical nervous systems did.

We should be investigating what is special about electrochemical computation.

Computational AI versus subcomputational biology

In discussions of artificial intelligence (AI), it is often assumed that implementing computations of a certain kind is sufficient for (or both necessary and sufficient for) 'phenomenal consciousness' (hereafter **consciousness**) (see Glossary). For example, an influential paper on AI consciousness with 19 authors (including a recipient of the 2018 Turing Award, the 'Nobel Prize of computing') announces that their 'method for investigating whether current or near-future AI systems might be conscious is based on three assumptions' [1]; the first of these is '**computational functionalism**' defined as: 'Implementing computations of a certain kind is necessary and sufficient for consciousness…'.

I argue that current discussions do not take sufficiently seriously the possibility that implementing certain computations is not sufficient because there could also be a **subcomputational** biologically necessary condition (or necessary and sufficient condition). This unspecified subcomputational biologically necessary condition – to be clarified later – is what I refer to as the **meat** condition.

The perspective I will develop is that when deciding whether a non-human system is conscious, we have no choice but to extrapolate from ourselves. We should have the most confidence in the extrapolations that consider both our nature from the perspective of computational **functionalism** and our nature from the perspective of subcomputational biology. However, we may soon encounter AI systems that resemble us only in their computational functional aspects, just as we already have many animal systems that share certain features of our underlying biology but lack the computations characteristic of human cognition. If we focus on computational similarities, we may doubt the consciousness of animals such as insects; if we focus on subcomputational biology, we may doubt the consciousness of AI. In this way, our criteria for evaluating consciousness in AI and animals can lead us to opposite conclusions. That said, people's beliefs about consciousness will likely depend primarily on factors such as whether the systems in question have a face and resemble us behaviorally [2].

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In what follows, I briefly explain the computation/subcomputation distinction and relate it to a more general distinction between functional role and realizer or implementer of that role. Then, focusing on what meat might be, and why it might be necessary for consciousness, I discuss the emergence in evolutionary history of the first animals and what was special about some of them.

An important preliminary: consciousness can have a function even if functionalism is false. Consciousness might have a function only because it has a meat implementation. For example, suppose the functions of consciousness are to motivate us to seek pleasure or avoid pain. Consciousness might have these functions in us only because these functions have a meat implementation in us. Perhaps primitive forms of consciousness arose by chance and evolution harnessed them to do these things, further amplifying their conscious properties for more effective motivation. Perhaps a non-conscious AI could use non-conscious states for motivational purposes.

Role versus realizer

I begin with the contrast between computational properties and subcomputational properties. The contrast is most easily understood when considering the simplest computational entities, logic gates. An AND gate is defined by the fact that two 1 inputs yield a 1 output, and all other combinations of 1 and 0 inputs yield a 0 output. That computational property can be realized in different ways, for example, electrically or mechanically. Both are illustrated in Figure 1.

In terms of this distinction, we may assume that our conscious states have certain computational properties, but of course, they also must have subcomputational properties in order to have computational properties. We do not know whether our consciousness depends directly on the computational, on the subcomputational properties, or both. The meat hypothesis is that our consciousness does depend directly on the subcomputational properties, even if it also depends on the computational properties.

It will be helpful to consider a more general form of the computational/subcomputational distinction in terms of causal roles and their **realizers** [3–5]. A causal role is a network of causal relations. To illustrate, consider a simplified example of a causal role, namely a causal disposition. Dormitivity (say of a pill) can be thought of as the disposition to cause sleep (when ingested). Fragility is the disposition for certain events – say being hit with a hammer – to cause breaking. Glass is fragile because its amorphous molecular structure does not allow it to bend without breaking. By contrast, cast iron is fragile because of micro-cracks. Here, fragility is a simple causal role, and amorphous structure and micro-cracks are the internal structures that implement that role.

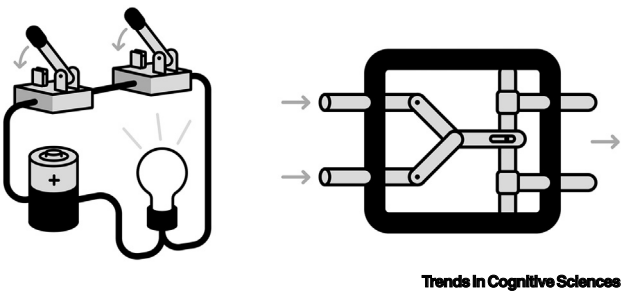

Figure 1. Illustration of computation versus subcomputation. These two devices are computationally equivalent because they are both AND gates. In an AND gate, two 1 inputs yield a 1 output and all other inputs yield a 0 output. Left: the closed switches are 1 inputs, whereas the light being on is a 1 output. Both switches being closed causes the light to go on, but any other combination of switches on or off does not. That makes the device an AND gate. Right: a mechanical AND gate in which a 1 input is pushing one of the dowels on the left and a 1 output is the extension of both dowels on the right. Both left dowels must be pushed to get the dowels on the right to move. These two devices are equivalent computationally but differ subcomputationally, just as a hypothetical AI of the near future might be computationally equivalent to a human but differ subcomputationally. Thanks to Tim Peacock for the drawing.

Glossary

Cnidaria: a phylum that includes true jellyfish, sea anemones, corals, hydra, and other species.
Computational functionalism: as applied to consciousness, the view that implementing certain computations is sufficient for consciousness, or, as defined by some, both necessary and sufficient.
Consciousness: as used in this article, consciousness is phenomenal consciousness, the subjectivity of experience.
Ctenophora: a phylum composed of comb jellies.
Electrochemical processing: electrical signals propagate within neurons, but the neurons communicate via chemical neurotransmitters.
First order property: a property that is not defined in terms of a role involving a cluster of other first order properties.
Functionalism: the view that implementing certain functions is sufficient for consciousness.
Functional role: causal relations among sensory inputs, motor outputs, and internal states.
Meat: the biological realizer of the functional role of consciousness.
Myriazoa: the clade containing sponges and humans but not comb jellies.
Porifera: sponges.
Realizer: the realizer of a functional role is the state that implements the role. A distinction is often made between a core realizer and a total realizer. The core realizer is what distinguishes one phenomenal character from another, for example, the experience of red from the experience of green. The total realizer includes enough of the background properties so as to be sufficient for a conscious experience. I refer here to core realizers.
Second order property: a property whose definition 'quantifies over' first order properties, that is, a property defined in terms of relations among a cluster of first order properties. A first order property is not so defined.
Sisters: sister groups are the most closely related groups sharing a common ancestor from which they diverged at roughly the same time.
Subcomputational properties: realizers of computational properties.

The role in dormitivity and fragility is a simple input–output process, but the more interesting roles also have an internal component. Thus, we could think of the pain role as involving certain sensory inputs (e.g., sharp things striking bodily surfaces), certain motor outputs (e.g., wincing), but also certain internal causal relations, for example, pains tend to cause thoughts of calling the doctor.

Roles can be understood as '**second order**' **properties** [6]. They are specified in terms of clusters of **first order properties** that are related in a certain way, whereas first order properties are not defined in that way. For example, dormitivity can be thought of as the second order property (e.g., of a pill) of possessing some first order property that causes sleep. A sample first order property that suffices is the chemical property $C_{12}H_{18}N_2NaO_3$ (secobarbitol). Often, there will be many first order properties that can realize or implement a given role. For example, $C_{16}H_{13}ClN_2O$ (diazepam) can also implement the dormitivity role. **Functional role** properties form a subset of second order properties – they consist in a causal role in regard to sensory inputs, motor outputs, and other first order properties that mediate between the inputs and outputs. Only some of these functional properties are computational [7]. The second/first order distinction helps to see that the two extrapolation bases belong to different ontological categories.

In these terms, the main point of this paper is that we have excellent reason to believe that certain combinations of (second order) functional roles – such as the ones present in us – together with their first order realizers are associated with consciousness. But that does not tell us whether consciousness depends directly on the functional roles, on the realizers that implement those roles independently of the roles they realize, or both. (Of course, if the roles are directly responsible, then the realizers are derivatively responsible because they are the underpinnings of the roles.)

To hedge our bets, we may extrapolate from both second and first order properties in deciding difficult cases. In thinking about infant consciousness (e.g., [8,9]) researchers have cited both roles and realizers. Infants are subject to visual masking and the attentional blink, both properties that are part of the details of roles. And they also show aspects of brain connectivity, which is partly defined in terms of realizers.

The hard cases, though, arise when there is a disparity between the evidence involving realizers and roles. We may soon have AI that has our functional roles with very different realizers. And there are cognitively simple animals that share some of the biological mechanisms that realize consciousness in us, yet lack the cognitive sophistication necessary for our roles. In these terms, the problem with functionalism is that it focuses on the roles, neglecting the realizers. It is cognitive sophistication that motivates ascribing consciousness to AI, but the reaction of fish to opiates involves both realizers and roles [10].

In terms of the role/realizer distinction then, meat can be defined as the biological realizer of our conscious roles, the first order implementations of our second order properties. And in those terms, we do not know from our own consciousness whether it is the meat or the role that is crucial for our consciousness.

When I speak of a realizer, I am talking about realizing mechanisms, not just material composition, not just realizing stuff. My view might be called 'subcomputationalism'. For example, I will suggest later that fluctuations in the ionic composition of the neurotransmitter soup between neurons might be an important mechanism of consciousness. But the specific ions used in extant organisms (sodium, potassium, calcium) are not necessary for this mechanism – other ions could function in this way.

The role and the realizer hypotheses are equally plausible, but most major theories of consciousness assume functionalism, that is, the role hypothesis (Box 1). The main reason is that it is easier to study what consciousness does than what it is – that is why David Chalmers called the former the 'easy problems' and the latter 'the hard problem' [11].

The role hypothesis may appear explanatorily superior because it seems to explain why the meat is important to consciousness: meat is important in us according to the role hypothesis because it fulfills a specific information processing role. When we investigate what aspects of our biology are important to consciousness, we focus on those that carry out the information processing roles that we think consciousness has. How can this practice be understood without supposing that the information processing roles are what make the biological state conscious?

From the point of view of the meat hypothesis, functionalism has it backwards: it is biologically grounded consciousness that is in part responsible for the information processing roles. The functionalists are saying that what consciousness is and what it does amount to the same thing. While the meat hypothesis acknowledges that the information processing role is important, specifically for the functions consciousness performs, it maintains that consciousness itself is something different from the role, based in biology. It may appear simpler to collapse what consciousness does and what it is into one thing, but that simplicity is illusory if that role can be accomplished by something other than biology (which may not be possible, as Rosa Cao points out [12]) or if consciousness might not have that role in some circumstances.

A further set of issues arises because it may be alleged that the meat hypothesis leads to the Hard Problem [11]: even if we were to find out that activation in the fusiform face area underlies our

Box 1. Prominent theories of consciousness are meat-neutral forms of functionalism

The most prominent theories of consciousness include (i) Global workspace theory (GWT), (ii) Higher order pointer theory (HOP), (iii) Local recurrence theory, and (iv) Integrated Information Theory (IIT). All these theories allow for both meat and non-meat implementations and are, in that sense, meat-neutral.

According to GWT [32], automatically activated perceptual processors (mainly in sensory areas) trigger activations in perceptual circuits that compete with one another for dominance. A few of these activations win, and some trigger self-reinforcing neural coalitions involving cognitive processing in fronto-parietal circuits. This process, 'ignition', makes the perceptual information available to cognitive mechanisms, and that process of making representations available to cognition is global broadcasting, the essence of consciousness. Global broadcasting can be characterized completely computationally and so does not require meat. GWT is a form of computational functionalism.

Higher order theories of consciousness hold that a perception is conscious if it is accompanied by another state of the same system that represents the perception. The dominant higher order theory is the 'pointer' account in which the cognitive representation of the perceptual state is a pointer that does not have perceptual content [33,34]. Since a pointer is a computational entity, this theory is a version of computational functionalism, and so it does not require meat.

The recurrent processing view [35] is not really a Theory (with a capital 'T') of consciousness since it restricts itself to vision and also provides a rather minimal necessary condition, emphasizing the importance of feedback loops to consciousness. Recurrent processing in this sense is a simple computational requirement, and so the doctrine that recurrent processing is required is a meat-free form of computational functionalism.

IIT is difficult to summarize without using its proprietary terminology, which is not always clear to others (but see [36,37]). IIT measures the degree of consciousness of a system by a quantity ('phi') that quantifies both how complex it is and how that complexity is internally integrated. IIT is not a version of computational functionalism in the sense of the term that applies to the other major players because, according to IIT, consciousness does not require any actual computation or indeed any activation of any circuits. But it is functionalist in a weaker sense because according to IIT, whether a system is conscious depends on structures that have certain counterfactual roles, that is, they determine what their states would cause if certain activations happened. Any mechanism can be conscious according to this theory if its states have the structures that have those counterfactual properties, independently of whether the mechanisms that realize it involve meat.

experience of faces, there would still be the question of why that activation is the basis of that phenomenal experience as opposed to others or none. There is an 'explanatory gap' between the brain activation and the experience [13].

But the idea that functionalism – the role hypothesis – solves the Hard Problem and closes the explanatory gap is a mirage. It only solves the Hard Problem if you think that all there is to explain about conscious experience is its functional role and not 'what it is like' to experience it [14].

In sum, focusing on role leads to one picture of consciousness, and on realizer leads to a different picture: the first favors AI, the second favors consciousness in simple animals.

But are realizers not computational too?

Given the success of computational models in biology, it makes sense to ask whether computation is fundamental or just a way of modeling and describing something more fundamental.

A simple, convincing, and correct – but superficial – answer is that computing the gravitational attraction of one body on another does not produce any gravitational force. The computations involved model gravitation but do not constitute any kind of gravitational or other force. The point is clear if we assume Newtonian gravitation: the force of one object on another is equal to the gravitational constant G times the masses of each object divided by the square of the distance between them. You can compute that until the heat death of the universe without producing any gravitational force.

There are two reasons this answer is superficial. The minor one is that it can be alleged that further computations embodying the assumptions of post-Newtonian physics would produce gravity. I do not take this reply seriously, but the second reason is deeper: that not only is neuroscience computational, but biology itself is computational at a more fundamental level.

Replying effectively and less superficially to this objection introduces a tricky dialectic that I can only summarize briefly here [15]. The main point is that whether any level of science is computational can be criticized by moving down the hierarchy to a lower, more fundamental non-computational level. If one argues that those lower levels are also computational, the response is that eventually we reach a level of basic physics whose realizations may not be computational.

Some speculate that the entities and relations underneath basic physics are themselves computational or informational; this is a form of what is sometimes termed 'it from bit' [16]. However, unlike 'it from bit', which remains a highly controversial idea, my point here is much less contentious, or should be.

If the meat-neutral theories of consciousness are wrong and meat is necessary for consciousness, what is it about meat that makes it necessary? As Peter Godfrey-Smith notes [17], 'Fully fledged conscious experience probably evolved from marginal and gray-area forms in neurally simpler animals, through a series of situations where the trait was neither clearly present nor absent….'. There is no reason to assume that there was a 'lights on' moment. If we favor gradualism in evolution, one place to look would be at the evolutionary origins of consciousness. One approach would be to consider the first animals, a matter to which I now turn.

The first animals

The two candidates for the earliest animals are **Ctenophora** (comb jellies) and **Porifera** (sponges).

There is an obvious reason to favor sponges as the first animals: they lack a nervous system, muscles, and a gut, whereas comb jellies have all three. If comb jellies were the earliest animals, it would suggest that sponges were their descendants, and if so, that sponges mysteriously lost those properties. The two hypotheses are illustrated in Figure 2. The hypothesis that comb jellies were first is labeled in the figure as the 'comb jelly **sister** hypothesis', with the word 'sister' indicating that the comb jellies of today may have diverged from a common ancestor they share with the sister to comb jellies of 600 million years ago. (I will refer to the sister of comb jellies as comb jellies except when disambiguation is needed.) The 'sponge sister hypothesis' is also depicted in Figure 2. Note that comb jellies appear twice in the diagram, illustrating their two possible positions in evolutionary history.

In 2023, the hypothesis that comb jellies (ctenophores) were the first animals received a huge boost in an article in *Nature* entitled 'Ancient gene linkages support ctenophores as sister to all other animals' [18]. This article demonstrated that comb jelly chromosomes were radically

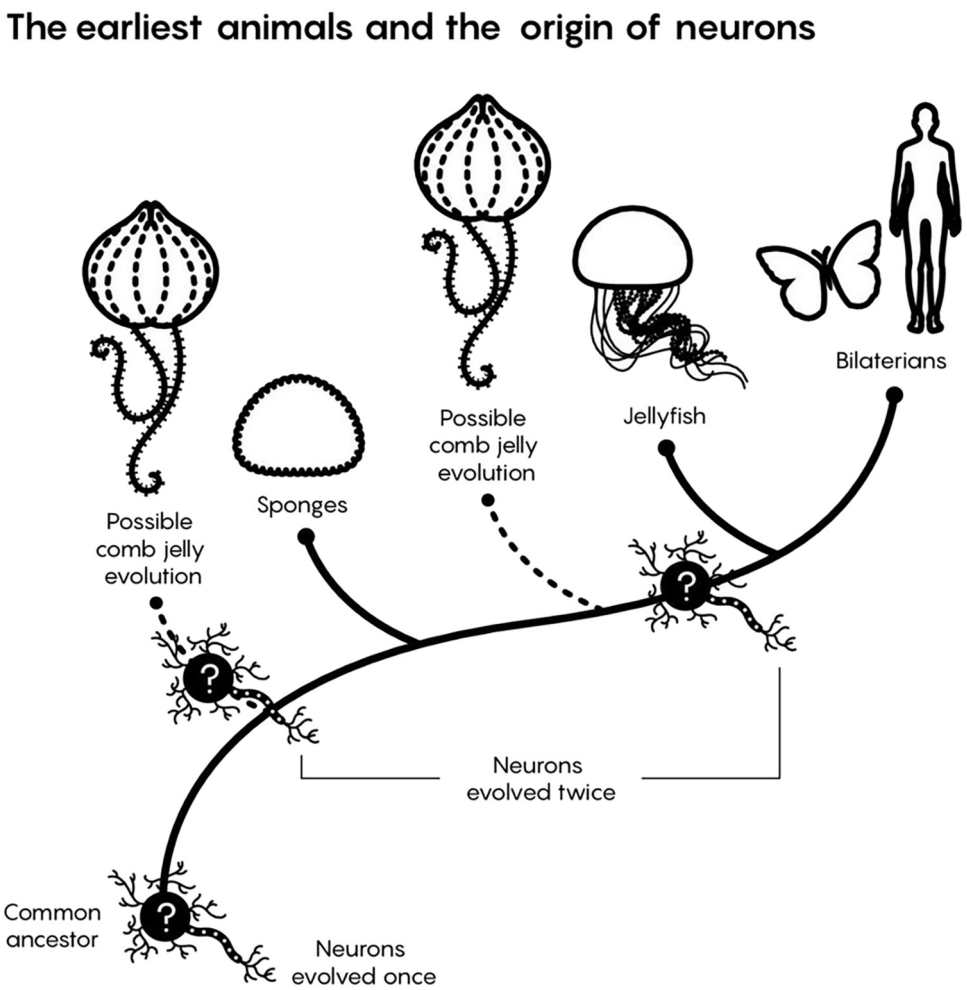

Figure 2. The origin of animals. The two hypotheses about the origin of animals, the comb jelly sister hypothesis and the sponge sister hypothesis. Note that comb jellies appear twice, once as the sister of all animals and again as part of the sponge sister hypothesis. From E. Singer (2015) Did neurons evolve twice? in Quanta Magazine[iii], by Olena Shmahalo with permission of Quanta Magazine.

different from those of other animals, including both sponges and humans, and further that the chromosomal divergence was 'the result of a rare and irreversible chromosome fusion-and-mixing event that provides robust and unambiguous phylogenetic support for the ctenophore-sister hypothesis' (page 110). The authors coined the term '**myriazoa**' for the clade containing sponges, humans, and all other animals except ctenophores.

However, if sponges did lose their neurons, one wonders why. A discovery that may help to answer the question was published in *Science*, also in 2023 [19], and widely discussed in many popular venues[i]. As the headline in *Science News* put it: 'Comb jellies have a bizarre nervous system unlike any other animal'[ii]. What this study showed is that in at least one stage of the comb jelly life cycle, they have an entirely electrical nervous system except at points of interface with the environment, where chemical synapses connect to sensory transducers and motor effectors.

This fact presents us with three salient evolutionary hypotheses. First, the ancestor of all animals might have had a primitive form of neurons. In this scenario, these primitive neurons could have evolved into the electrical nervous system seen in the comb jelly and the electrochemical neurons found in other animals, including us, while being lost in the sponges. In this scenario, neurons evolved once and were then modified or lost in different lineages. Second, the electrical neurons of comb jellies could have later developed into our electrochemical neurons. In the third scenario, the ancestor to comb jellies was the progenitor to sponges, losing their neurons, with neurons evolving again. But in all these scenarios, it appears that **electrochemical processing** has a functional advantage over purely electrical processing.

What is meant by electrochemical mechanisms? In myriazoans like us, electrical signals propagate within neurons, but most communication between neurons occurs via chemical signals. The chemical part of our nervous systems involves the release of neurotransmitters from the axon of the firing neuron into the chemical soup (extracellular fluid) that separates it from the dendrites of neighboring neurons. These neurotransmitters are then taken up by the receptors on these dendrites, which cause ion channels to open, allowing sodium intake and generating an electrical signal flowing down the axons (the action potential) of the receiving neurons. This may result in these neurons firing and another release of neurotransmitters. Our nervous systems have a number of kinds of neural connections, including a minority of direct electrical connections between neurons (gap junctions).

If there is something valuable about the electrochemical kind of nervous system we have, including any potential link to consciousness, what would it be? One possibility is neural inhibition. As I explain in Box 2, efficient electrical inhibition circuits are possible, so this may not be the main advantage.

What else might be important about electrochemical processing? An obvious candidate is learning. We can approach this possibility by asking whether the electrochemical processing in the more complex organisms that evolved from sponges and comb jellies enables a new or enhanced form of learning. The best candidate is **Cnidaria**, a phylum that includes true jellyfish, sea anemones, corals, hydra, and others. What is the evidence of learning in these animals? A recent review of learning in Cnidaria [20] showed sensitization in which an initial stimulus (e.g., a prey creature brushing a sensor) makes the cell more sensitive and short-term habituation in which a repeatedly stimulated cell decreases responses. But these are within-cell non-associative phenomena (i.e., they do not involve a relatively permanent response that is specific to the stimulus type) and have been observed in plants, fungi, and single-celled organisms [21].

There are two kinds of associative learning: classical conditioning and operant conditioning. Classical conditioning involves the association of two stimuli, as famously demonstrated when

Box 2. Is inhibition the major advantage of electrochemical nervous systems?

Inhibition is as important in the neural system as activation. Inhibition is essential to the operation of the visual system since it enhances contrast changes, edge detection, and direction sensitivity. In addition, inhibition is crucial to attention (and indirectly, to consciousness) since attention only boosts some signals by inhibiting others [38–40]. Neural inhibition seems an obvious candidate for a function of chemical synapses because it may seem that jolts of electricity could activate but not inhibit.

Early in the twentieth century it was widely agreed that the nervous system had nerve cells, the 'neuron doctrine'. Although the effect of chemicals on neurons was well known, the dominant view – held largely by neurophysiologists – was that neural transmission was electrical, though there were 'pharmacologists' who favored chemical signaling. The *War of the Soups and the Sparks* (as Elliot Valenstein called it [41]) chronicles this dispute. Otto Loewi showed in 1921 that what we now call acetylcholine (a neurotransmitter) slowed the beating of a frog heart, but the electrical picture still held sway in part because even after the activation of the heart by the vagus nerve was shown to be chemical, chemical transmission of signals was regarded as too slow for operation in the brain.

The key player on both sides was J.C. Eccles. In 1947, Brookes and Eccles [42] showed that there were plausible electrical models of inhibition. However, in the 1950s, Eccles showed that an electrochemical model of the major kind of synapse predicted some phenomena better than the purely electrical model. Later work has shown that electrical synapses enable all sorts of inhibitory effects that are important in neural computation [43] and in generating inhibitory rhythms. Electrical and electrochemical inhibition each have their advantages. Electrical inhibition is faster than chemical inhibition and can be bidirectional. And electrical synapses are more reliable than chemical synapses. In any case, the relevance to our discussion of what is so special about electrochemical processing is that, since inhibition may and is done efficiently electrically, the evolutionary importance of electrochemical processing may not be its use in inhibition.

Pavlov's dogs came to salivate in response to the bell that had been paired with food. Operant conditioning is more sophisticated – it involves a voluntary response, often occurring by chance, which is followed by reward or punishment. As a result, the voluntary response becomes more or less frequent, depending on positive or negative reinforcement.

The review of learning in Cnidaria [20] identified only one well-controlled test that supported classical conditioning and none that 'firmly supported' operant conditioning. (The limited evidence on learning in comb jellies is mixed [22,23].) There is one cnidarian that is capable of operant conditioning, the box jellyfish, but it appears to have evolved a hundred million years after the most primitive cnidarians [24].

Furthermore, by many measures, comb jellies are more complex organisms, both behaviorally and physiologically, than the early cnidarians that followed them, excluding later cnidarians like the box jellyfish. Comb jellies have a more complex body, including a distinct mouth and anus, whereas cnidarians typically have a single opening that serves both functions. Comb jellies also have a sophisticated propulsion system with multiple comb rows and a balance organ, whereas early cnidarians were stationary polyps anchored to solid material (similar to coral and sea anemones) and had not yet evolved eyes or medusae. The one cell type in which cnidarians were more complex was the nematocysts for stinging prey, which were more complex than the comb jellies' sticky colloblasts that served the same purpose [25,26].

Chemical synapses confer a number of advantages in complex organisms, but it is not clear that these advantages would have played a role in the initial evolution of chemical synapses. For example, chemical synapses have a major role in brain rhythms. In many advanced animals, axons have so many synapses that post-synaptic neural firing requires rhythmic action of populations of neurons, and in these animals, waves of ion fluctuations in the chemical soup surrounding the neurons are important [27]. See Box 3 on one aspect of brain rhythms.

So where does that leave us? A purely electrical nervous system has not led to the same degree of sophistication as an electrochemical system, but we do not know why electrochemical neural

Box 3. What role do large-scale dynamic patterns play in consciousness?

Another candidate for the biological basis of consciousness, proposed by Godfrey-Smith [17,44], is the presence of large-scale dynamic brain patterns. This view fits well with the electrochemical mechanisms featured in the text because rhythmic activity in the chemical soup outside of neurons is important to driving brain waves, and we know that anesthesia and sleep involve changes in large-scale dynamics.

Godfrey-Smith motivates the relevance of these patterns, especially with regard to the unity of experience, concluding [17] '…the combination of network properties and large-scale dynamic patterns seen in nervous systems is important in the explanation of consciousness'.

However, some data suggest that at least some large-scale dynamic patterns have little psychological significance in vision. In 1990, Francis Crick and Christof Koch suggested that 40 Hz brain oscillations were important to binding conscious features together into a unified experience [45]. Many people pointed out that alternating current oscillates at 60 Hz in the USA and 50 Hz in Europe, and that oscillating current causes flickering in some electric lights, which in turn have an oscillatory effect on the brain. Specifically, fluorescent lights have substantial flicker, as did the early LED lights. However, fluorescent lights are not known to cause disruptions in consciousness.

Crick and Koch's suggestion was put to the test in some experiments. As noted in one paper [46] (page 539), 'Cortical neurons respond to stimulus flicker with a rhythmic, phase-locked discharge …and many cells respond to temporal modulation at frequencies in the 'oscillation' range…The firing elicited by flicker is similar to the synchronous oscillatory activity of cortical neurons, which is said to underlie perceptual grouping phenomena'. The authors of this paper examined perceptual grouping with and without flicker, concluding that flicker does not affect the visual phenomena they examined. They conclude (page 543), 'The lack of any effect of this flicker or of its phase suggests that the temporal coherence of oscillatory cortical responses may be unrelated to the processes that link perceptual features across a scene'. Godfrey-Smith is concerned with a broader kind of unity of experience, involving interoception, affect, and perception, but this result does motivate further investigation of what the difference is between large-scale dynamic patterns that are and are not relevant to consciousness.

processing originally developed or why it proved to be more successful. We have reason (admittedly, only weak reason) to doubt that either inhibition or learning was the initial trigger for the evolution of an electrochemical nervous system. See Outstanding questions for a brief recital of what we do not know.

Concluding remarks

Suppose electrochemical processing engendered a glimmer of what Seth calls the 'inchoate, shapeless formless feeling' of being alive [28]. What was the evolutionary advantage of that glimmer? We have reason to think that conscious processing can accomplish in adult humans what our unconscious processing cannot [29]. But even if this is right, what could a mere glimmer of conscious activation accomplish? Perhaps it offered an advantage unrelated to consciousness, but it is also possible that the predominance of this kind of processing is a lucky accident, one that, hundreds of millions of years later, provided the foundation for our conscious experience.

Another possibility is that the 'meat' in meat machines is not directly relevant to consciousness but instead enables functions that themselves are the basis of consciousness. If this is the case, AI that replicates our functional organization would be a much better candidate for consciousness than simple animals that share our biology.

In any case, the main argument of this paper, presented in the section on roles versus realizers, is that when deciding whether machines or animals are conscious, we have to extrapolate from the only creatures we know to be conscious: namely, us. Extrapolating requires deciding what properties of the extrapolation base are relevant. These properties fall into two general categories: second order functional roles and the first order realizers that implement those roles. If we focus on the functional roles, we will be inclined to ascribe consciousness to cognitively sophisticated systems, potentially including AI systems. If we focus instead on the realizers that implement those roles, we could potentially decide that very simple cognitively unsophisticated biological creatures are conscious. And in that sense, there is an opposition between AI and simple animals.

Outstanding questions

How can we tell whether the roles, the realizers (or both) are responsible for our conscious experience? Do AI systems that have not been designed to mimic us (unlike current large language models) show markers of consciousness? Do naturally evolved creatures whose ecological niche is so different from ours that their functional roles bear little resemblance to our own evince markers of consciousness?

If the realizers are important, what is it about them that makes the difference, and how would we tell? Perhaps a sophisticated organic creature could be designed or simulated that had only electrical synapses. Would it show markers of consciousness?

If we can determine whether meat is necessary for consciousness, would that discovery bring us closer to a solution to the Hard Problem?

What are the prospects for building an artificial system that has electrochemical properties like ours?

If, contrary to the assumption of this article, 'illusionism' is true, can we still ask whether it is the roles or the realizers of the roles that are responsible for phenomenal consciousness? Illusionism about consciousness – made famous by Daniel Dennett – says that there is no such thing as phenomenal consciousness as distinct from access-consciousness, the global availability of information.

If we develop robots that look, act, and talk like humans, it will be tempting to treat them as conscious, and if so, does that push us toward a disjunctive theory of consciousness, one that specifies that consciousness can either be grounded in functional role or in biology?

There is little doubt that some aspect of our biology is necessary for consciousness in us, and if necessary in us, might be necessary, period. While this is a modest conclusion, it is important if true and therefore warrants serious consideration. In practice, we will inevitably be most sure of the consciousness of systems that are like us both computationally and subcomputationally, and less sure when one or the other is absent.

Note that even if consciousness does require meat, that leaves open the possibility of AI suffering, because suffering may not require consciousness. Indeed, the stifling of AI agency might itself be a kind of suffering [30]. I doubt that current AI models have anything that could be called agency, but future AI systems could conceivably have agency without consciousness. Arguably, humans can act in a way that involves unconscious agency. For example, some have argued that the dorsal visual system functions as something like a robot that is given abstract instructions but has some independence about how to implement them [31].

If meat is directly responsible for consciousness, we do not know what biological feature is responsible or why. A breakthrough will be needed for progress, and failing to recognize that a breakthrough is needed may not be conducive to finding it.

Resources

[i] www.scientificamerican.com/article/did-neurons-evolve-twice/

[ii] J. Buehler, *Science News* (2023) 203, 000–000

[iii] www.quantamagazine.org/comb-jelly-neurons-spark-evolution-debate-20150325/

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